Data schema — AwkBeats Prep Station

Document control

Field	Value
Project	AwkBeats Prep Station
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Author	Cursor Agent, for owner Awkward J
Repo	https://github.com/Awkward8787/awkbeats-prep-station
Branch	`cursor/awkbeats-prep-station-e2e2`
Commit SHA	f99d3c6a38ad256e123fa13b50e0f7660369bd61
Live URL	None. Data stays on the producer machine.
Operator	Awkward J (legal entity UNCONFIRMED)

Table of contents

1. Data architecture
2. Entity diagram
3. Entities
4. Client storage
5. IPC reference
6. Third-party API calls
7. Webhooks
8. Background jobs
9. Files
10. Lifecycle
11. Migrations and seeding
12. Security of data
13. Glossary
14. Open questions
15. Changelog
16. Sources

1. Data architecture

There is no database server. Three places hold data:

Data	Where
Catalog and settings	`library.json` in the Electron userData directory
Cover images	`userData/artwork/<beat id>.jpg`
Audio	The path the producer imported. Not copied.

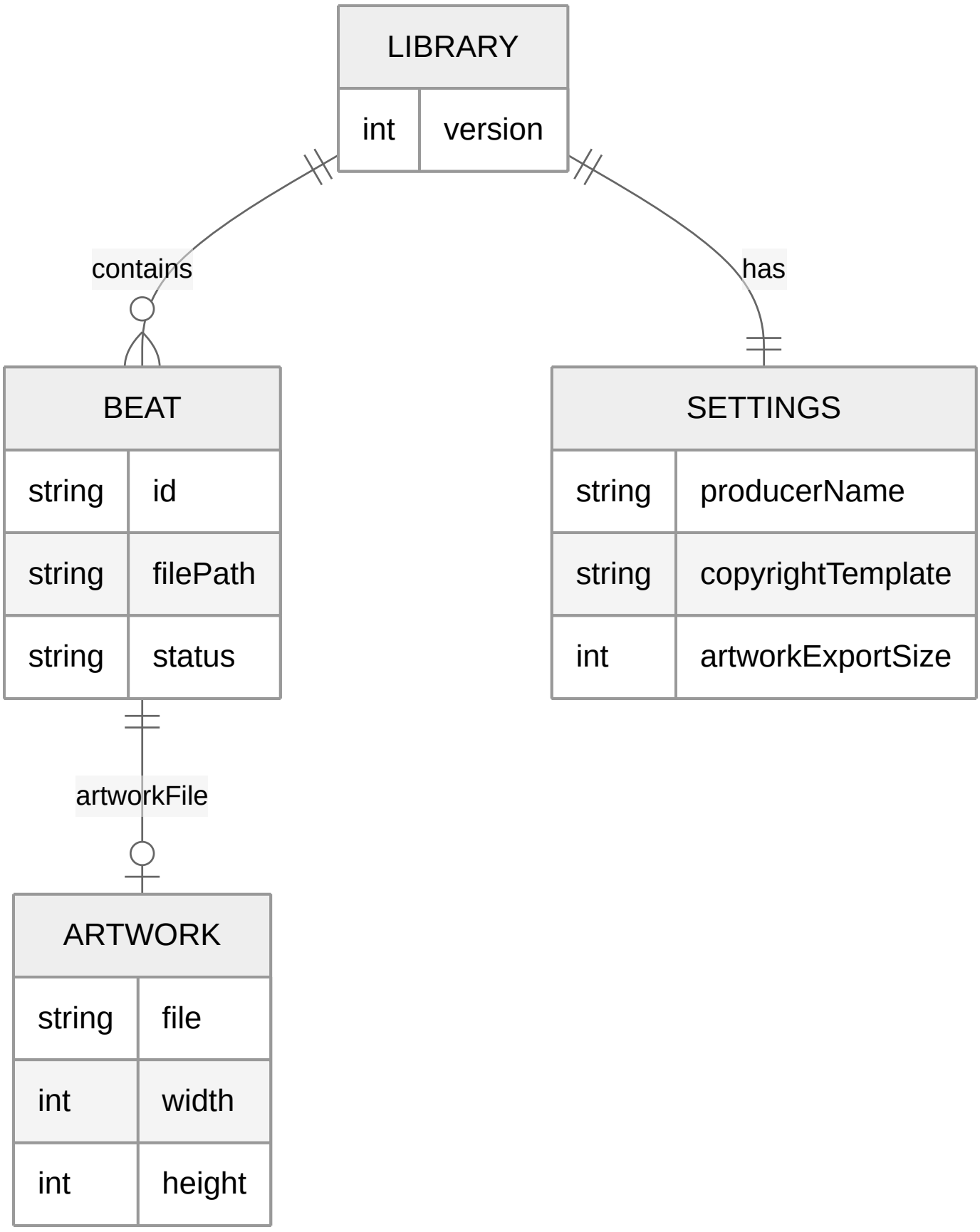
Data	Where
UI state (selection, filters, sort)	React memory. Lost on quit.



The JSON catalog points at audio paths and at artwork files. The window's filters are not persisted.

Override userData with `AWKBEATS_USER_DATA`. The default path is Electron's for this app id (`com.awkbeats.prepstation`).

2. Entity diagram



One library file holds many beats and one settings object. Artwork is a sibling file, not a row inside the JSON.

3. Entities

`missing` is computed on read (`fs.existsSync`) and is not a reason to treat the file as a second source of truth. It is still present on the TypeScript type. `LibraryStore.save` writes the in-memory beats, which include whatever `missing` value they had when loaded. Callers should treat `missing` from `list()` as live and the stored boolean as stale. That is a known quirk (debt, low): the next save can persist the previous `missing` flag. `list()` overwrites it before return.

LibraryFile

Field	Type	Nullable	Default	Constraints	Description	Example	PII
version	number	no	1	must be 1 or load throws	Schema version	1	no
beats	Beat[]	no	[]	array	Catalog	[]	mixed
settings	Settings	no	DEFAULT_SETTINGS	object	App settings	see below	no

Settings

Field	Type	Nullable	Default	Constraints	Description	Example	PII
producerName	string	no	Awkward J	none	Written as artist fallback and as PRODUCER tag	Awkward J	name
copyrightTemplate	string	no	© {year} Awkward J	`{year}` replaced at import	Default copyright	© 2026 Awkward J	no
artworkExportSize	1500 or 3000	no	3000	only those two accepted in `updateSettings`	Default crop export	3000	no

Beat

Implemented in `src/core/library/types.ts`. `id` is `crypto.randomUUID()` from `BeatRecord.create`.

Field	Type	Nullable	Default	Constraints	Description	Example	PII
id	string	no	uuid	unique in practice	Primary id	a uuid	no
filePath	string	no	—	unique per library (`findByPath`)	Absolute audio path	/Users/me/beats/Night Drive.wav	path may contain a username
fileName	string	no	basename	—	Display name	Night Drive.wav	no
format	wav, mp3, aiff, flac	no	—	enum	Container	wav	no
fileSize	number	no	stat size	bytes, import max 2 GiB	Size	1440004	no
durationSec	number	yes	null	seconds	Duration	8.0	no
sampleRate	number	yes	null	Hz	Rate	44100	no
title	string	no	tag or basename	—	Title	Night Drive	no
artist	string	no	tag or producerName	—	Producer / artist	Awkward J	name
genre	string	no	""	free text, suggestions in `BeatCatalog.genres`	Genre	Trap	no

Field	Type	Nullable	Default	Constraints	Description	Example	PII
mood	string	no	""	suggestions in `BeatCatalog.moods`	Mood	Dark	no
tags	string[]	no	[]	max 24, each max 32, case-insensitive unique	Tags	night, keys	no
comment	string	no	""	max 4000 on update	Comment	—	could be personal
copyright	string	no	template	max 400 on update	Copyright line	© 2026 Awkward J	no
isrc	string	no	""	empty or 12-char canonical	ISRC	USAWK2600001	no
date	string	no	tag or mtime	stored as YYYY-MM-DD slice	Date	2026-10-21	no
bpmDetected	number	yes	null	0.01 resolution when set	Detector BPM	140.25	no
bpmScale	0.5, 1, or 2	no	1	enum	Half or double	1	no
bpmOverride	number	yes	null	40–300 if set	Hand BPM	null	no
bpmLocked	boolean	no	false	—	Lock	false	no
bpmConfidence	number	yes	null	0–1	Detector confidence	0.98	no
keyDetected	string	yes	null	Camelot name	Detected key	F# minor	no
camelotDetected	string	yes	null	like 11A	Detected code	11A	no
keyOverride	string	yes	null	—	Hand key	null	no
camelotOverride	string	yes	null	—	Hand code	null	no
keyLocked	boolean	no	false	—	Lock	false	no
keyConfidence	number	yes	null	0–1	Gap score	1	no
status	BeatStatus	no	idea	idea, in_progress, ready, leased, sold	Catalog status	ready	no
notes	string	no	""	—	Internal notes, not written to the audio tag	Hold for Friday	could be personal
artworkFile	string	yes	null	filename only	Cover filename	`<id>.jpg`	no
artworkWidth	number	yes	null	pixels	Width	1800	no
artworkHeight	number	yes	null	pixels	Height	2400	no
peaks	number[]	no	[]	160 values, 3 decimals, 0–1	Waveform	0.12	no
missing	boolean	no	false	recomputed on read	File absent	false	no
analyzedAt	string	yes	null	ISO timestamp	Last analysis	2026-10-21T00:00:00.000Z	no
createdAt	string	no	now	ISO	Created	same	no
updatedAt	string	no	now	ISO	Updated	same	no
importedAt	string	no	now	ISO	Imported	same	no

Enums

Beat status labels (`BeatStatusLabel`): Idea, In progress, Ready, Leased, Sold.

Genre suggestions: Trap, Drill, Hip-Hop, R&B, Pop, Afrobeats, Amapiano, Jersey Club, House, Techno, Boom Bap, Lo-fi, Rage, Dancehall, Reggaeton. The field accepts other strings.

Mood suggestions: Dark, Energetic, Chill, Melodic, Aggressive, Sad, Bouncy, Soulful, Spacey, Uplifting.

Key names come from `Camelot.roots`: C, C#, D, D#, E, F, F#, G, G#, A, A#, B, plus `major` or `minor`. Flats parse onto the sharp spelling (`Eb major` is `D# major`).

Tag payload written into audio

`TagCommit.fromBeat` maps the working BPM and working key, plus `settings.producerName` as `producer`. ID3 frames (`TagWriter.id3Tags`):

App field	ID3	Vorbis	Notes
title	TIT2	TITLE	
artist	TPE1	ARTIST	
genre	TCON	GENRE	
comment	COMM eng	COMMENT	
bpm	TBPM with two decimals	BPM	
key	TKEY up to 3 chars (`F#m`) plus TXXX KEY	KEY and INITIALKEY (Camelot)	TKEY cannot hold "F# minor"
camelot	TXXX CAMELOT	CAMELOT	
tags	TXXX TAGS, comma-separated	TAGS	
mood	TXXX MOOD	MOOD	
copyright	TCOP if it can be made to start with a year, plus TXXX COPYRIGHT	COPYRIGHT	`TagText.copyrightFrame`
isrc	TSRC	ISRC	
date	TDRL and TYER when the first four chars are a year	DATE	
producer	TXXX PRODUCER	PRODUCER	from settings, not the beat artist field
artwork	APIC type 3	PICTURE type 3	

WAV also writes RIFF LIST/INFO (INAM, IART, IGNR, ICMT, ICRD, ICOP). Existing INFO and `id3 ` chunks are replaced. Other RIFF chunks, including `data`, stay. AIFF writes NAME, AUTH, ANNO, and ID3, replacing older copies of those chunks.

FLAC keeps STREAMINFO and audio frames, drops old VORBIS_COMMENT (type 4), PICTURE (type 6), and PADDING (type 1), then writes a new comment, an optional picture, and 256 bytes of padding. Vendor string: `AwkBeats Prep Station`.

CSV columns

`CsvExport.headers`: id, title, artist, file_name, file_path, format, duration_sec, bpm, bpm_detected, bpm_scale, bpm_override, bpm_locked, key, camelot, key_detected, camelot_detected, key_locked, key_confidence, genre, mood, tags (joined with `|`), status, notes, comment, copyright, isrc, date, analyzed_at, imported_at, updated_at. RFC 4180 quoting.

4. Client-side storage

localStorage, sessionStorage, IndexedDB, and cookies are not used.

The only persisted client state is `library.json` and the artwork directory, written by the main process. Shape: `LibraryFile` above. Version strategy: `load()` throws if `version !== 1`. There is no migrator yet. Size: one JSON rewrite of the full catalog per change. Cleared by deleting the userData folder, or by Remove for individual beats (artwork file deleted, audio kept).

5. IPC reference

These are not HTTP endpoints. They are `ipcMain.handle` channels. Auth is the local user. No rate limit. The preload method is the name in `PrepApi`.

ID	Channel	Preload	Purpose	Payload	Success	Error	Side effects	File
API-001	library:list	list	All beats	none	Beat[]	throw on bad JSON at load	recomputes missing	main.ts
API-002	settings:get	settings	Read settings	none	Settings	—	none	main.ts
API-003	settings:update	updateSettings	Patch settings	Partial Settings	Settings	ignores illegal export size	writes JSON	LibraryStore
API-004	library:import	importPaths	Import	string[] paths	ImportResult	per-file messages inside result	writes JSON, may write artwork	Importer
API-005	dialog:audio	pickAudio	Open files	none	string[]	[] if cancelled	none	main.ts
API-006	dialog:folder	pickFolder	Open folder	none	string or null	null if cancelled	none	main.ts
API-007	dialog:artwork	pickArtwork	Open image	none	{name, bytes} or null	null if cancelled	none	main.ts
API-008	beat:update	updateBeat	Patch one beat	id, UserPatch	{beat} or {error}	error string, no throw for validation	writes JSON	LibraryStore
API-009	beat:batch	batchUpdate	Patch many	ids, BatchPatch	Beat[]	—	writes JSON	LibraryStore
API-010	beat:remove	remove	Remove rows	ids	Beat[]	—	deletes artwork files, writes JSON	LibraryStore
API-011	beat:analyze	analyze	Analyze WAV/AIFF	ids	{beats, needsDecode, errors}	errors[] for failures	writes JSON	main.ts
API-012	beat:analysis	submitAnalysis	Store renderer analysis	id, AnalysisResult	{beat} or {error}	—	writes JSON, respects locks	LibraryStore.applyAnalysis
API-013	beat:audio	readAudio	Whole file bytes	id	Uint8Array	throws if missing	none	main.ts
API-014	beat:write-tags	writeTags	Write audio tags	ids	{id, ok, message}[]	message on failure	mutates the audio file	TagWriter
API-015	beat:artwork	saveArtwork	Store JPEG	id, bytes, width, height	Beat	—	writes jpg and JSON	LibraryStore
API-016	beat:export-artwork	exportArtwork	Save dialog	id	path or null	null if none	writes a copy	main.ts
API-017	library:csv	exportCsv	Save CSV	ids (empty means all)	path or null	null if cancelled	writes the CSV the user chose	CsvExport
API-018	beat:reveal	showInFolder	Show in folder	id	void	—	OS shell	shell.showItemInFolder

ID	Channel	Preload	Purpose	Payload	Success	Error	Side effects	File
API-019	app:site	openSite	Open brand site	none	void	—	external browser	https://awkbeats.com only
API-020	app:startup	takeStartup	MP3/FLAC imported via `--import`	none	string[] ids, then cleared	—	none after read	main.ts
API-021	menu-command	onMenu	Menu to renderer	command string	event	—	none	main.ts
API-022	beat://audio/id and beat://cover/id	audio and img src	Stream file	id in URL	file bytes	empty if unknown	none	protocol.handle

Example `beat:update` validation error body: `{ "error": "BPM must be between 40 and 300." }`.

`--import <path>` on the executable runs import before the window loads and queues compressed ids for `app:startup`.

Idempotency: import of an existing path is a duplicate skip. Analyze overwrites unlocked analysis fields. Tag write replaces prior ID3, INFO, or Vorbis comment blocks.

6. Third-party API calls made by the server

Not applicable. The main process does not call a remote API. `shell.openExternal('https://awkbeats.com')` is a browser handoff, not an API call. music-metadata and node-id3 are in-process.

7. Webhooks

Not applicable. None incoming or outgoing.

8. Background jobs and scheduled tasks

Not applicable. No cron, no queue. Analysis runs on the IPC call stack (main for WAV/AIFF, renderer for MP3/FLAC). A long file is capped at 90 seconds of samples inside `TrackAnalyzer`.

9. File and object storage

Path	Permissions	Contents	Cleanup
userData/library.json	user umask	Catalog	Replaced atomically via `library.json.tmp` then rename
userData/artwork/*.jpg	user umask	Cover	Deleted in `LibraryStore.remove`
Audio path on the beat	unchanged	Original audio	Never deleted by the app
CSV path from the save dialog	user choice	Export	Not tracked
Exported cover from the save dialog	user choice	JPEG copy	Not tracked

No sweep of old files. Orphan artwork can remain if a beat id is removed from the JSON by hand. The app's remove path deletes the jpg it knows about.

10. Data lifecycle

Retention: until the producer deletes the files. No backup is taken by the app. Restore: copy `library.json` and `artwork/` back into userData. Audio must still be at `filePath`. If a path changed, the row shows missing; v1 has no relink dialog.

Deletion: Remove drops the row and the artwork. It does not delete audio. Uninstalling the app does not delete userData on every OS; Electron's default is to leave it. The producer deletes that folder to wipe the catalog.

11. Migrations and seeding

No migration runner. A future version must bump `version` and teach `load()` to upgrade. Rollback of a bad JSON is the `.tmp` file only if a crash happens before rename; after rename, the previous file is gone. Keep a copy if you experiment.

Seeding: `scripts/seed-demo.ts` is a development tool. It is not invoked by the app or by CI. It writes generated loops, tags, and a cover into a directory you pass. Do not point it at a real catalog. `tests/fixtures/silence.flac` is a 0.1 second silent FLAC used only by tests.

12. Security of data

No encryption at rest beyond what the operating system does for the user account. No passwords, no hashes. Access control is the OS user. The renderer cannot read `library.json` except through IPC. Tag writes require a library id, and the path written is the path stored on that beat, not a path chosen by the renderer in the write call.

`beat:audio` returns the full file. That is intended for local decode. Do not expose the preload bridge to untrusted web content. Navigation guards in `electron/main.ts` exist for that reason.

13. Glossary

Term	Meaning
userData	Electron application data directory.
Working BPM	`bpmOverride` else `bpmDetected * bpmScale`.
TXXX	ID3 user-text frame. Used for MOOD, TAGS, KEY, CAMELOT, PRODUCER, COPYRIGHT.
APIC	ID3 attached picture.

14. Open questions

ID	Question	Owner	Blocked
Q-003	Validate ISRC against a real registrant prefix?	Awkward J	Stricter than the 12-character pattern
Q-011	Should notes be written into the audio comment as well?	Awkward J	v1 keeps notes only in the catalog
Q-012	Relink when a file moves?	Awkward J	v1 marks the row missing

15. Changelog

Date	Version	Change	Why
2026-10-21	1.0	Schema for library version 1 and the IPC list.	Owner build brief.

16. Sources

- src/core/library/types.ts, LibraryStore.ts, CsvExport.ts, Importer.ts, AudioFiles.ts, BeatRules.ts.
- src/core/tags/TagWriter.ts, TagCommit.ts, TagTypes.ts, TagReader.ts.
- electron/main.ts, electron/preload.ts.